Tae Yong Song

Website | Google Scholar | GitHub | LinkedIn

Seoul, Republic of Korea

Ph.D. Student in Building Science
Department of Architecture and Architectural Engineering
Seoul National University

RESEARCH PROFILE

I study building intelligence at the intersection of building physics, performance simulation, and artificial intelligence. My work focuses on models that remain physically meaningful, transferable, and useful for building-energy decisions when operational data are limited.

EDUCATION

2026–present **Ph.D. in Architecture and Architectural Engineering**, Seoul National University

Building science and building performance simulation

2022–2023 **M.S. in Engineering**, Seoul National University

Thesis: "Exploring Potential of Transfer Learning in Overcoming Data Scarcity for Building Energy Model"

2015–2021 **B.S. in Architecture and Architectural Engineering**, Seoul National University

Graduated in 2021

RESEARCH INTERESTS

- Physics-embedded and physics-informed machine learning
- Building energy and thermal performance
- Building performance simulation and reduced-order modeling
- Data-efficient learning, transfer learning, and digital twins

PUBLICATIONS

International Journal Article

2026 Physics-embedded hybrid modelling approach for room temperature prediction using Siamese neural network and RC model

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, Junu Lee, and Cheol-Soo Park. *Journal of Building Performance Simulation*, 19(2), 277–288.

International Conference Papers

2026 Curated Intelligence for Visual Inference and Deterministic Logic toward Automatic BEM

Song, T. Y., Kim, J. H., Lee, S. J., Kim, S. K., Park, H. C., Kim, K. J., and Park, C. S. *Proceedings of the 7th Building Simulation Applications Conference*, June 24–26, Bozen-Bolzano, Italy.

2024 Physics-Informed Hybrid Modeling Approach for Room Temperature Prediction Using an RC Model and Siamese Neural Network

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, and Cheol-Soo Park. *SimBuild 2024*.

2023 Causality-based model for decision-making of building energy retrofit

Chul-Hong Park, Young-Sub Kim, **Tae Yong Song**, Seon-Jung Ra, and Cheol-Soo Park. *Building Simulation 2023*, 1433–1438.

2023 Limitations of Conventional Artificial Neural Network-Based Surrogate Models for Building Energy Retrofit

Chul-Hong Park, Young-Sub Kim, Seon-Jung Ra, **Tae Yong Song**, and Cheol-Soo Park. *ASHRAE Annual Conference*, 683–690.

Domestic Conference Papers

2026 BIM to BEM 자동화를 위한 맞춤형 AI 기술 개발: LLM, 컴퓨터 비전, 들루네 기하 알고리즘, 최소순환 알고리즘 통합

송태용, 김진홍, 김수경, 박현철, 김경재, 박철수. 대한건축학회 춘계학술발표대회논문집, 제46권 제1호, 829--830.

2022 NODE-GAM을 통한 동적 민감도 분석

송태용, 박철홍, 라선중, 김영섭, 박철수. 한국건축친환경설비학회 추계학술발표대회 논문집, 181--182.

RESEARCH PROJECTS

2026–2030 **Virtual Building Modeling and Simulation Framework, and a Causality-Based AR Tool for Net-Zero Building Design Evaluation**

Korea Institute of Civil Engineering and Building Technology (KICT)
Contributing to a virtual-building simulation framework, causal performance evaluation, and AR-based communication of net-zero design outcomes.

2025–2026 **LLM Technology for Digital-Twin-Based Building Energy Models**

Samsung Electronics
Developed LLM-assisted workflows for organizing, generating, and interpreting building-energy model information in a digital-twin environment.

2025–2026 **IDF Model Development for Digital-Twin Energy Simulation**

Eterion (Samsung Electronics)
Developed and reviewed simulation-ready EnergyPlus IDF models and model-validation workflows for building-energy digital twins.

2022–2024 **Automated Digital Diagnosis and Design Technologies for Green Remodeling**

Korea Agency for Infrastructure Technology Advancement (KAIA)
Contributed to building-energy modeling, data-driven performance diagnosis, and automated evaluation of green-remodeling alternatives.

2023 **Data-Driven HVAC Load Prediction and Optimal Air-Handling-Unit Control**

LG Electronics
Developed and evaluated HVAC load-prediction models for control-oriented air-handling-unit operation.

2022–2023 **Greenhouse-Gas Reduction Factors and Emissions Assessment for Building Green Remodeling**

Ministry of Land, Infrastructure and Transport (MOLIT)
Supported energy simulation and scenario analysis for estimating greenhouse-gas reduction factors under alternative retrofit conditions.

TEACHING EXPERIENCE

2026–present **Teaching Assistant**, Seoul National University

Department of Architecture and Architectural Engineering

2023 **Teaching Assistant**, Seoul National University

Department of Architecture and Architectural Engineering

HONOR

2026 **Outstanding Presentation Paper Award**

Architectural Institute of Korea Spring Annual Conference, Oral Presentation – General Division

Last updated: July 2026 Full publication details and project updates: taeyong-song.github.io